v2.0 — HC-SR04 + ST7735 + MAX98357A + inductive proximity

Branch: `v2.0` (based on the ST7735 display branch)

Download & flash

Prebuilt **v2.0 ST7735** firmware (`FW_VERSION 2.0.0`):

File	Use
[`../bin/v2.0/DetektorAir-v2.0-ST7735-factory.bin`](../bin/v2.0/DetektorAir-v2.0-ST7735-factory.bin)	new / blank board, flash at **0x0**
[`../bin/v2.0/DetektorAir-v2.0-ST7735-ota.bin`](../bin/v2.0/DetektorAir-v2.0-ST7735-ota.bin)	web **CONFIG → UPDATE FIRMWARE** or ArduinoOTA

Blank board over USB (run from the repository root):

```cmd
esptool.py --chip esp32 --port COM5 write_flash 0x0
bin\v2.0\DetektorAir-v2.0-ST7735-factory.bin
```

Existing v2.0 board: upload `bin/v2.0/DetektorAir-v2.0-ST7735-ota.bin` from the web CONFIG tab (or use ArduinoOTA on port 3232, password `lacakota`).

Checksums: [`../bin/v2.0/SHA256SUMS.txt`](../bin/v2.0/SHA256SUMS.txt).

> The older images are in [`../bin/legacy/`](../bin/legacy/):
> `DetektorAir-ST7735-factory.bin` and `DetektorAir-ST7735-ota.bin` are from the **servo sweep** ST7735 branch. Do not use them for v2.0.

Renders

ST7735 boot	ST7735 idle	ST7735 detected
		





Wiring (exact)

> **Common ground is mandatory.** Connect the ESP32 GND, HC-SR04 GND, ST7735 GND, MAX98357A GND, 5 V supply 0 V and inductive-sensor 0 V together.

Power

Recommended supply: **5 V, at least 2 A** (WiFi peaks + audio transients).

From	To	Note
5 V PSU `+`	ESP32 `5V` / `VIN`	do not use the 3.3 V pin
5 V PSU `+`	HC-SR04 `VCC`	HC-SR04 is a 5 V module
5 V PSU `+`	MAX98357A `VIN`	amplifier supply
5 V PSU `−`	common `GND`	star ground
ESP32 `3V3`	ST7735 `VCC`	display logic is 3.3 V
ESP32 `3V3`	MAX98357A `SD`	enables the amplifier
ESP32 `3V3`	10 kΩ pull-up	inductive sensor output

Add **470 µF + 100 nF** across 5 V/GND near the MAX98357A. Do **not** power the HC-SR04 or MAX98357A from the ESP32 3.3 V regulator.

> If you power the ESP32 through its `5V` pin, disconnect USB first (or use USB as the only 5 V source) on boards that do not have a power-path diode.

ESP32 board (38-pin DevKit)

This build targets a standard **ESP32 DevKit 38-pin** board (ESP32-WROOM-32, 4 MB flash) **without a microSD slot**. Use the **GPIO labels printed on the**

board**, not the physical pin position, because 30-pin and 38-pin boards place the same GPIOs differently.

- `5V` / `VIN` → 5 V supply; `3V3` → ST7735 VCC; `GND` → common ground.
- GPIO 34 is input-only (inductive sensor). GPIO 35/36/39 are also input-only.
- GPIO 6–11 are connected to the SPI flash and must not be used.
- GPIO 5 is a strapping pin but is used for HC-SR04 `TRIG`; it is high-impedance at boot, so it is normally safe.
- Some 38-pin boards have a **microSD slot** wired to GPIO 2/4/12/13/14/15. This build's board has **no SD slot**, so GPIO 13/14 are free for the ST7735. If you use a different 38-pin board that does have the slot, remove any SD card and verify that the display initialises; otherwise remap the TFT pins and rebuild.
- The onboard LED is usually GPIO 2; it is not used.

Complete pin map

Module	Module pin	Connects to	Wire / note
HC-SR04	`VCC`	5 V	
	`TRIG`	**GPIO 5**	direct, 3.3 V trigger
	`ECHO`	**GPIO 18**	through 1 kΩ / 2 kΩ divider
	`GND`	GND	
ST7735	`VCC`	3.3 V	
	`GND`	GND	
	`SCL` / `SCLK`	**GPIO 14**	SPI clock
	`SDA` / `MOSI`	**GPIO 13**	SPI data
	`CS`	**GPIO 27**	
	`DC` / `A0`	**GPIO 26**	
	`RES` / `RST`	**GPIO 25**	
	`LED` / `BL`	**GPIO 32**	HIGH = backlight on
MAX98357A	`VIN`	5 V	
	`GND`	GND	
	`BCLK`	**GPIO 19**	I2S bit clock
	`LRC` / `LRCLK`	**GPIO 21**	I2S word clock
	`DIN`	**GPIO 22**	I2S data
	`SD`	3.3 V	amplifier enable
	`GAIN`	leave open	default 9 dB
	`OUT+`	speaker `+`	4 Ω or 8 Ω
	`OUT−`	speaker `−`	never to GND
Inductive sensor	brown	+6–36 V sensor supply	not ESP32 5 V unless rated
	blue	0 V / GND	common with ESP32
	black	**GPIO 34** via 1 kΩ	NPN NO output
Pull-up	10 kΩ	GPIO 34 → 3.3 V	mandatory
Reset button	leg 1	**GPIO 33**	
	leg 2	GND	momentary NO to GND

HC-SR04

```text
HC-SR04 VCC  ──────────────── 5 V
HC-SR04 GND  ──────────────── GND
HC-SR04 TRIG ──────────────── GPIO 5
HC-SR04 ECHO ── 1 kΩ ─┬────── GPIO 18
                      └─── 2 kΩ ─── GND
```

- `ECHO` is a **5 V** output. The 1 kΩ / 2 kΩ divider gives about **3.3 V** at GPIO 18. Never connect ECHO directly to the ESP32.
- `TRIG` accepts the ESP32 3.3 V pulse.
- GPIO 5 is a boot strapping pin. It works with the HC-SR04 `TRIG` input (high impedance), but if the board has trouble booting, move `TRIG` to GPIO 4 and rebuild.
- Range 2–400 cm. The HC-SR04 is **not waterproof**; keep the transducer dry.
- Aim the sensor at the contact/water surface and keep the cone clear.

ST7735 display

```text
ST7735 VCC ──── 3.3 V
ST7735 GND ──── GND
ST7735 SCL ──── GPIO 14
```

```
ST7735 SDA ──────── GPIO 13
ST7735 CS  ──────── GPIO 27
ST7735 DC  ──────── GPIO 26
ST7735 RES ──────── GPIO 25
ST7735 LED ──────── GPIO 32
```

- Use the **128×160 SPI, no-touch** module. MISO, touch and microSD are unused.
- Firmware is configured as **160×128 landscape**, `ST7735_INITB`, `TFT_RGB_ORDER = TFT_BGR`, rotation `1`.
- `LED`/`BL` is driven HIGH by GPIO 32. If the module's backlight draws more than about 12 mA, drive it with a 2N2222/MOSFET instead of the GPIO directly.
- If colours are swapped or the image is offset, change `ST7735_INITB` / `TFT_RGB_ORDER` / `ST7735_ROTATION` in `firmware/ESP32_Radar/st7735_ui.h`.

Confirmed 1.8" ST7735S module

This is the module specified for v2.0: 1.8 inch, 128×160, SPI, ST7735S, 3.3 V, 12 o'clock viewing direction.

Header pin	Label	Connect to
1	`GND`	GND
2	`VCC`	**3.3 V**
3	`SCL`	**GPIO 14**
4	`SDA`	**GPIO 13**
5	`RES`	**GPIO 25**
6	`DC`	**GPIO 26**
7	`CS`	**GPIO 27**
8	`BL`	**GPIO 32**

`BL` is the backlight-control pin. If the panel stays dark while GPIO 32 is HIGH, the module's `BL` is active-low: set `TFT_BACKLIGHT_ON` to `LOW` in `platformio.ini` and `st7735_ui.h`.

Module compatibility

This build targets the common **1.8 inch, 128×160, SPI, non-touch ST7735** module (ST7735S / ST7735R, 8 pins).

- Native resolution is 128×160. The firmware draws **160×128 landscape** (`ST7735_ROTATION 1`).
- The standard 8-pin module is `VCC GND SCL SDA RES DC CS BL/LED` and matches the pin map above exactly.
- **Not** compatible with 1.44" 128×128, 0.96"/1.14" 80×160, 135×240, or 160×80 ST7735 modules; the layout is fixed at 160×128.
- Touch or microSD variants are fine: leave `T_CLK`, `T_CS`, `T_DIN`, `T_DO`, `T_IRQ` and the SD-card pins unconnected. The firmware uses neither.
- The blue 8-pin **ST7735S** board labeled `VCC GND SCL SDA RST DC CS BLK` matches this build. Because it is an ST7735S (7735R family), if the default `ST7735_INITB` gives a blank, offset or odd-colour image, try `ST7735_REDTAB` first, then `ST7735_GREENTAB`.
- Some boards use `BLK` as the backlight **cathode**. If the screen stays dark with GPIO 32 HIGH, set `TFT_BACKLIGHT_ON` to `LOW`, or tie `BLK` to GND.
- If the image is shifted by 1–3 px or the colours are wrong, select the init that matches your module in **both** `platformio.ini` and `st7735_ui.h`:
  - `ST7735_INITB` — current default (7735B-style 128×160)
  - `ST7735_GREENTAB` — 128×160, offset 2/1
  - `ST7735_GREENTAB2` — 128×160, offset 2/1, different MADCTL
  - `ST7735_GREENTAB3` — 128×160, offset 2/3
  - `ST7735_REDTAB` — 128×160 red-tab
- To swap red/blue, change `TFT_RGB_ORDER=TFT_BGR` to `TFT_RGB`.
- If the image is upside down, change `ST7735_ROTATION` from `1` to `3`.

MAX98357A audio

```text
MAX98357A VIN  ──────── 5 V
MAX98357A GND  ──────── GND
MAX98357A BCLK ──────── GPIO 19
MAX98357A LRC  ──────── GPIO 21
MAX98357A DIN  ──────── GPIO 22
```

```
MAX98357A SD   ──────── 3.3 V
MAX98357A GAIN ──────── leave open (9 dB)
MAX98357A OUT+ ──────── speaker +
MAX98357A OUT- ──────── speaker -
```

- Use a **4 Ω or 8 Ω speaker rated 3 W or more**.
- Never connect either speaker terminal to GND; the outputs are bridged.
- `SD` must be HIGH or the amplifier stays in shutdown.
- The firmware plays the embedded **MP3** alert on every inductive detection. To use the embedded WAV instead, set `PROX_AUDIO_USE_MP3 0` in `firmware/ESP32_Radar/audio_player.h`.
- Audio pins are set by `g_out->SetPinout(19, 21, 22);` in the same file.

LJ12A3-4-Z/BX inductive sensor (NPN NO)

```text
brown ──────── +6…36 V sensor supply
blue  ──────── 0 V / GND (common with ESP32)
black ── 1 kΩ ──┬──────── GPIO 34
                ├──────── 10 kΩ ──── 3.3 V
                └──────── 100 nF ─── GND   (optional noise filter)
```

- The NPN output is **HIGH when idle** and pulls **LOW** when metal is detected. This matches `INPUT` + external pull-up and NPN-NO logic.
- GPIO 34 is **input-only** and has no usable internal pull-up: the external **10 kΩ to 3.3 V is required**.
- Do **not** connect the black wire directly to the ESP32. It must go through the **1 kΩ series resistor**, with the **10 kΩ pull-up to 3.3 V** at the GPIO 34 node. Never feed the 6-36 V sensor supply into the ESP32.
- If the sensor uses a separate supply, its 0 V must still connect to ESP32 GND.
- Sensing distance is about **4 mm** (adjustable with the rear screw); the target must be metal.
- Firmware counts each new detection with **80 ms debounce** and saves the count in NVS.

Reset button

```text
GPIO 33 ──── momentary NO button ──── GND
```

- Internal pull-up is enabled in firmware; no external resistor is needed.
- Press for at least **40 ms**. This resets only the **proximity counter**, not the WiFi settings.

Pin restrictions

Pin	Reason
GPIO 6-11	connected to SPI flash on most ESP32 modules
GPIO 23	legacy buzzer assignment, not used in v2.0
GPIO 19 / 21 / 22	used by MAX98357A I2S audio
GPIO 34 / 35 / 36 / 39	input-only, no internal pull-up; GPIO 34 is the sensor input
GPIO 5	boot strapping pin; used by HC-SR04 TRIG (see above)

There is **no servo** in v2.0. The HC-SR04 measures continuously in one fixed direction.

Where the pins are defined

Signal	File
HC-SR04 TRIG / ECHO	`PIN_TRIG`, `PIN_ECHO` in `ESP32_Radar.ino`
ST7735 SPI + backlight	`platformio.ini` build flags **and** `st7735_ui.h`
MAX98357A I2S	`SetPinout(19, 21, 22)` in `audio_player.h`
Inductive sensor / button	`PIN_PROX`, `PIN_PROX_RESET` in `proximity.h`

> When changing ST7735 pins under PlatformIO, update **both** `platformio.ini`

> and `st7735_ui.h` so the PlatformIO and Arduino IDE builds stay identical.

Prebuilt display-init variants

The default image uses `ST7735_INITB`. If your ST7735S panel is blank, shifted, or has wrong colours, try one of these OTA images — same firmware, only the TFT init profile differs:

File	Init profile
[`../bin/v2.0/variants/DetektorAir-v2.0-ST7735S-redtab-ota.bin`](../bin/v2.0/variants/DetektorAir-v2.0-ST7735S-redtab-ota.bin)	`ST7735_REDTAB`
[`../bin/v2.0/variants/DetektorAir-v2.0-ST7735S-greentab-ota.bin`](../bin/v2.0/variants/DetektorAir-v2.0-ST7735S-greentab-ota.bin)	`ST7735_GREENTAB`

Upload them through **CONFIG → UPDATE FIRMWARE – OTA**, the same as the normal `-ota.bin`.

- Try **REDTAB** first for a normal 128×160 image.
- Try **GREENTAB** if REDTAB is shifted by 1–3 px or shows a stripe at the edge.
- If the default `INITB` image is already correct, no variant is needed.

Behaviour

- HC-SR04 measures continuously in one fixed direction.
- ST7735 shows the boot logo for 1.5 s, then distance and proximity count.
- The provided droplet artwork is an outline normally and fills with a green gradient when a contact is inside the alert threshold.
- Each new inductive detection increments and saves the counter in NVS.
- Reset the counter with the **physical button on GPIO33** or the **RESET COUNTER** button in the web UI.
- Each detection plays the embedded **MP3** alert through the MAX98357A. A WAV version is embedded too; switch with `PROX_AUDIO_USE_MP3`.
- Web UI and OTA remain available.
- Web background uses `web/bg.jpg`; the header logo remains `web/logo.jpg`.

Audio files

```text
firmware/ESP32_Radar/audio_prox.h   embedded MP3 + WAV arrays
firmware/ESP32_Radar/prox_alert.mp3 reference MP3
firmware/ESP32_Radar/prox_alert.wav reference WAV
```

ESP8266Audio is GPL-3.0; the firmware/source must be distributed together.

Build result

- PlatformIO: **1,274,177 bytes (64.8%)**, RAM **51,336 (15.7%)**
- Arduino: **1,460,491 bytes (74%)**, RAM **54,136 (16%)**
- Arduino IDE needs TFT_eSPI and ESP8266Audio 1.9.9 installed.